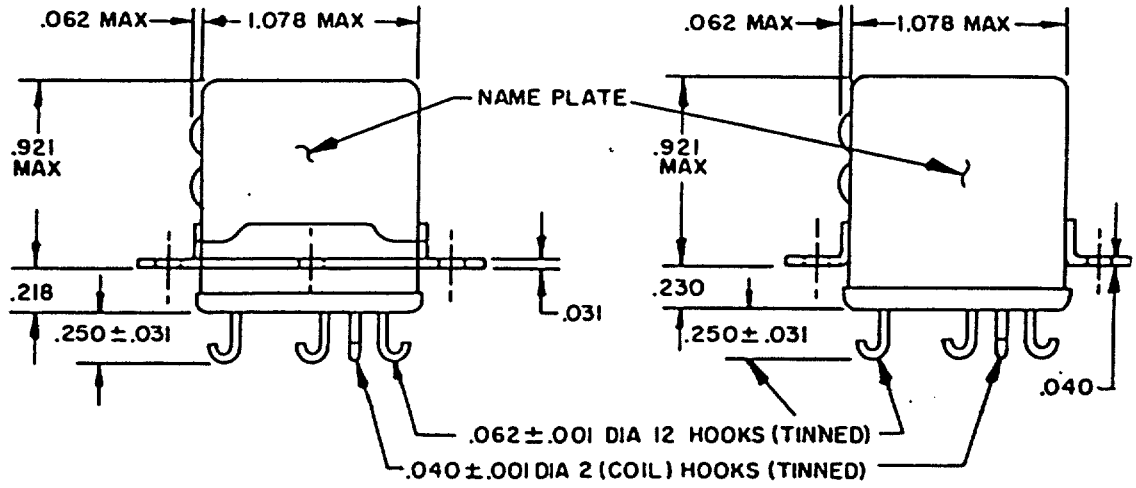
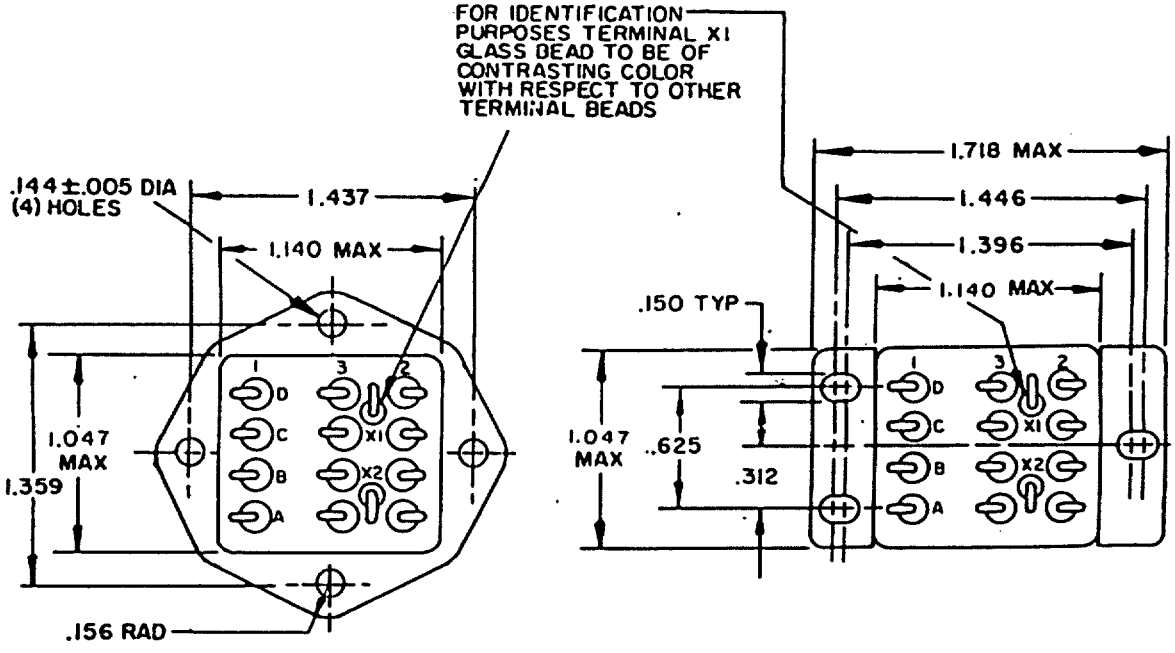
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FED. SUP CLASS
5945




DASH NUMBERS 1 AND 2

DASH NUMBERS 3 AND 4

(J) denotes changes

APPROVED 9 Nov 1964 REVISED (H) 5 JUN 87 (J) 10 FEB 89

User activities: Army - Navy - Air Force -

Review activities: Army - EC Navy - Air Force - 11, 99

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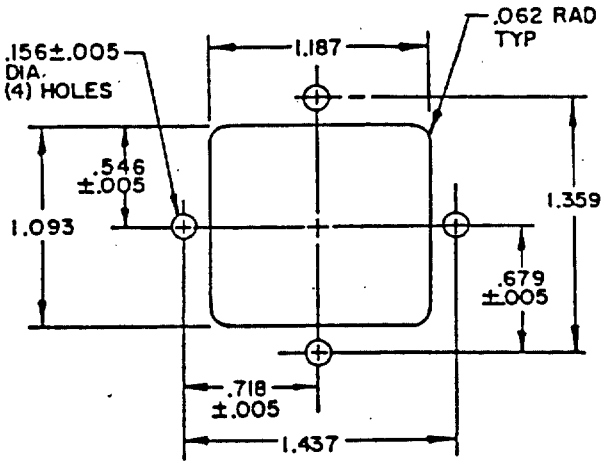
| P.A<br>USAF - 85<br>Other Cust<br>Navy - AS | International interest | TITLE<br>RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4PDT, TYPE I, SOLDER HOOKS HERMETICALLY SEALED | MILITARY STANDARD<br>MS27255 |
|---|---|---|---|
| Procurement Specification<br>MIL-R-6106 | | SUPERSEDES: | PAGE 1 OF 6 |

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AMSC N/A

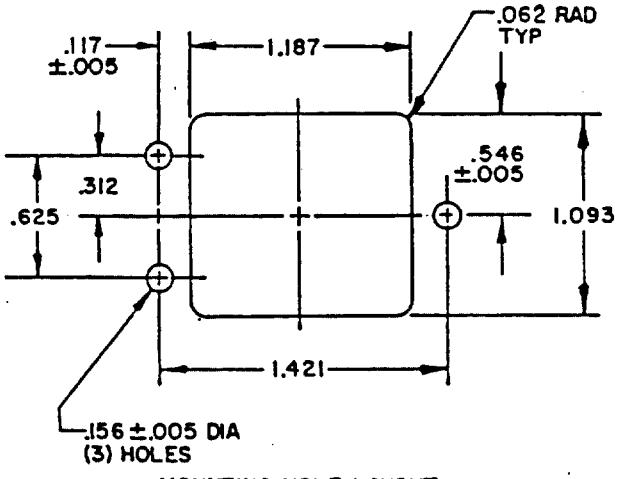
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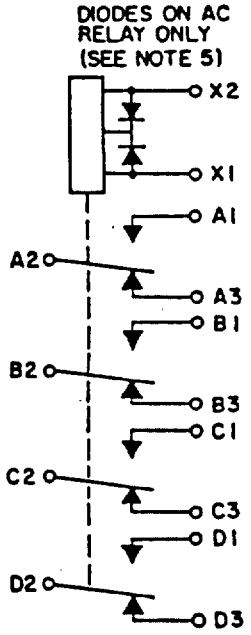

MOUNTING HOLE LAYOUT

## DASH NUMBERS 1 AND 2




MOUNTING HOLE LAYOUT

## DASH NUMBERS 3 AND 4




## CIRCUIT DIAGRAM

APPROVED 9 Nov 1964 REVISED (J) For changes see pages 4 and 6

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| | | | |
|---|---|---|---|
| P.A USAF - 85<br>Other Cust Navy - AS | International interest | TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4PDT, TYPE I, SOLDER HOOKS, HERMETICALLY SEALED | MILITARY STANDARD<br>MS27255 |
| Procurement Specification MIL-R-6106 | | SUPERSEDES: | PAGE 2 OF 6 |



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .001 | 0.03 | .218 | 5.54 | 1.047 | 26.59 |
| .005 | 0.13 | .230 | 5.84 | 1.078 | 27.38 |
| .031 | 0.79 | .250 | 6.35 | 1.093 | 27.76 |
| .040 | 1.02 | .312 | 7.92 | 1.140 | 28.96 |
| .062 | 1.57 | .315 | 13.87 | 1.187 | 30.15 |
| .117 | 2.97 | .546 | 15.88 | 1.359 | 34.52 |
| .144 | 3.66 | .679 | 17.25 | 1.396 | 35.46 |
| .150 | 3.81 | .718 | 18.24 | 1.437 | 36.50 |
| .156 | 3.96 | .921 | 23.39 | 1.446 | 36.73 |
| | | | | 1.718 | 43.64 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
7. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

| Dash number MS27255- | Type | Coil | Terminal type | Mounting or mating socket | Max weight in pounds |
|---|---|---|---|---|---|
| 1 | I | ac | Solder hook | Bracket | .20 |
| 2 | I | dc | Solder hook | Bracket | .20 |
| 3 | I | ac | Solder hook | Bracket | .20 |
| 4 | I | dc | Solder hook | Bracket | .20 |

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Review activities: Army - EC
Navy - EC
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

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Procurement Specification MIL-R-6106

P.A USAF - 85
Other Cust Navy - AS

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4PDT, TYPE I, SOLDER HOOKS, HERMETICALLY SEALED

MILITARY STANDARD

MS27255

PAGE 4 OF 6

FED. SUP CLASS 5945

(J) TABLE II. Operating characteristics.

| MS part no. MS27255 | Coil data | | | | | | | | | | | Time (milliseconds-maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Hold voltage 2/ | Drop-out voltage 2/ | Operate 3/ | Release 4/ | Contact bounce | | | |
| | | | | | | | | | | | | | | Main | | Aux | |
| | | Volts 1/ | Freq. Hz | Res Ω | Volts | Amperes | Normal 2/ | High temp test | Cont current test | | | | | NO | NC | NO | NC |
| -1 and -3 | ac | 115 | 400 | N/A | 120 | .04 | 90 | 95 | 103 | 30 | 5.0 | 15 | 30 | 3 | 5 | | |
| -2 and -4 | dc | 28 | N/A | 160 | 29 | .20 | 18 | 19.5 | 22.5 | 7.0 | 1.5 | 10 | 10 | 3 | 5 | | |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

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Review activities: Army -
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User activities: Army -
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TABLE III. Rated contact load (amperes per pole) (case grounded).

| Type of load | Life operating cycles X $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 Phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 100 | 10 | 10 | | | 10 | | | | 10 | | | | |
| Inductive | | | | | | 10 | | | | 10 | | | | |
| Inductive | 50 | 10 | 10 | | | | | | | | | | | |
| Motor | 100 | 5 | 5 | | | 5 | | | | 5 | | | | 2/ |
| Lamp | 100 | 3 | 3 | | | 3 | | | | 3 | | | | 2/ |
| Transfer load | | | | | | | | | | | | | | 3/ |
| Mechanical life reduced current | 400 | 2.5 | 2.5 | | | | | | | | | | | |
| Intmd current | Applicable per specifiction | | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ 100,000 operations - on NO contacts, and 50,000 operations on NC contacts at 28 V dc.
3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

P.A USAF - 85
Other Cust Navy - AS

International interest

Procurement Specification MIL-R-6106

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4PDT, TYPE I, SOLDER HOOKS, HERMETICALLY SEALED

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MS27255

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MILITARY STANDARD

MS27255

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## Environmental characteristics

| | |
|---|---|
| Temperature range | -70°C to +125°C |
| Max altitude rating | 80,000 ft |
| Shock G-level | 50 G |
| Duration | 11 ms |
| Max duration contact opening | 10 µs |
| Vibration - sinusoidal | |
| G-level | 20 G |
| Frequency range | 10 - 2,000 Hz |
| Vibration - random | |
| Applicable specification | N/A |
| Power spectral density | N/A |
| RMS G min | N/A |
| Frequency range | N/A |
| Curve | N/A |
| High shock | N/A |
| Acceleration | 15 G |

## Electrical characteristics

| | |
|---|---|
| Insulation resistance, initial | 100 megohms |
| After life or environmental tests | 50 megohms |

Dielectric strength (sea level)

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 V rms | 1,000 V rms |
| Aux contacts | | |
| All other points | 1,500 V rms | 1,000 V rms |

1/ Dielectric strength (altitude)

| | 80,000 ft |
|---|---|
| Coil to case | 250 V rms |
| Aux contacts | |
| All other points | 350 V rms |

| | |
|---|---|
| Max contact drop initial | 0.150 volt |
| After life test | 0.175 volt |
| Overload current | 40 amperes dc |
| Rupture current | 50 amperes dc |
| Duty rating | Continuous |
| RFI specification | MIL-STD-461 |

(Applicable to coil circuits of ac operated relays)

## Quality conformance inspection

Performance of groups B and C tests are not applicable.

1/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.

(J) Group A acceptance reports shall be submitted to the qualifying activity on a yearly basis in order to retain qualification for this military standard sheet.

FED. SUP CLASS 5945

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